

ML101

Lecture 04: Classification

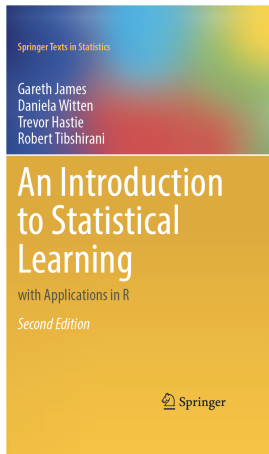
An Introduction to Statistical Learning

James, Witten, Hastie & Tibshirani

August 27, 2026

An Introduction to Statistical Learning with Applications in R

a.k.a. ISLR



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Visit <https://www.statlearning.com/>.

Plan

An Overview of Classification

Logistic Regression

Linear Discriminant Analysis

Classification Evaluation & Other Classifiers

Classification

- ▶ Qualitative variables take values in an unordered set $\mathcal{C}$, such as:

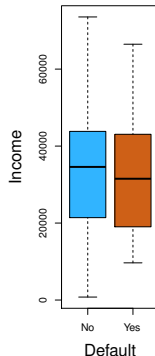
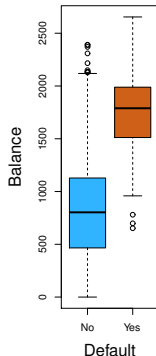
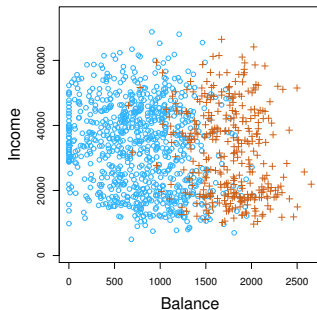
`eye color` $\in$ {brown, blue, green}
`email` $\in$ {spam, ham}.

- ▶ Given a feature vector X and a qualitative response Y taking values in the set $\mathcal{C}$, the classification task is to build a function $C(X)$ that takes as input the feature vector X and predicts its value for Y ; i.e. $C(X) \in \mathcal{C}$.
- ▶ Often we are more interested in estimating the *probabilities* that X belongs to each category in $\mathcal{C}$.
- ▶ For example, it is more valuable to have an estimate of the probability that an insurance claim is fraudulent, than a hard classification fraudulent or not.

Example: Credit Card Default

Default Data Set: Boxplots and scatterplots showing the relationship between **balance**, **income**, and credit card **default** status (No vs. Yes).

- ▶ Individuals who defaulted tended to have significantly higher credit card balances than those who did not.
- ▶ In contrast, income levels appear very similar between those who defaulted and those who did not.



Can we use Linear Regression?

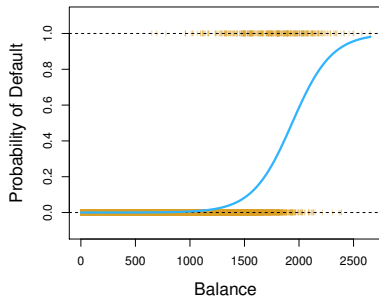
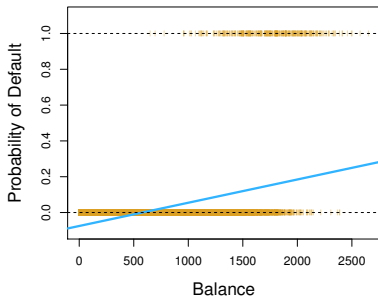
Suppose for the Default classification task that we code

$$Y = \begin{cases} 0 & \text{if No} \\ 1 & \text{if Yes.} \end{cases}$$

Can we simply perform a linear regression of Y on X and classify as Yes if $\hat{Y} > 0.5$?

- ▶ In this case of a binary outcome, linear regression does a good job as a classifier, and is equivalent to linear discriminant analysis which we discuss later.
- ▶ Since in the population $E(Y \mid X = x) = \Pr(Y = 1 \mid X = x)$, we might think that regression is perfect for this task.
- ▶ However, linear regression might produce probabilities less than zero or bigger than one. Logistic regression is more appropriate.

Linear versus Logistic Regression



Linear vs. Logistic Regression Fits for Binary Data:

- ▶ **Left (Linear Regression):** A straight regression line fitted to binary 0/1 data results in predicted probabilities that dip below 0.0 for low balances and exceed 1.0 for high balances.
- ▶ **Right (Logistic Regression):** The S-shaped logistic curve naturally asymptotes at 0.0 and 1.0, guaranteeing sensible probability estimates $\Pr(Y = 1 \mid X) \in [0, 1]$ everywhere.

Linear Regression continued

Now suppose we have a response variable with three possible values. A patient presents at the emergency room, and we must classify them according to their symptoms:

$$Y = \begin{cases} 1 & \text{if stroke;} \\ 2 & \text{if drug overdose;} \\ 3 & \text{if epileptic seizure.} \end{cases}$$

- ▶ This coding suggests an ordering, and in fact implies that the difference between stroke and drug overdose is the same as between drug overdose and epileptic seizure.
- ▶ Linear regression is not appropriate here.
- ▶ Multiclass Logistic Regression or Discriminant Analysis are more appropriate.

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Logistic Regression

Let's write $p(X) = \Pr(Y = 1 \mid X)$ for short and consider using **balance** to predict **default**. Logistic regression uses the form:

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}.$$

($e \approx 2.71828$ is Euler's number.)

- ▶ It is easy to see that no matter what values β_0, β_1 or X take, $p(X)$ will have values between 0 and 1.
- ▶ A bit of rearrangement gives:

$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X.$$

- ▶ This monotone transformation is called the **log odds** or **logit** transformation of $p(X)$ (by log we mean natural log: $\ln$).

Maximum Likelihood

We use **maximum likelihood** to estimate the parameters:

$$\ell(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i:y_i=0} (1 - p(x_i)).$$

- ▶ $p(x_i) = \Pr(Y = 1 \mid X = x_i)$: The predicted probability that individual i defaults.
- ▶ First Product ($\prod_{i:y_i=1} p(x_i)$): Multiplies the predicted probabilities for all individuals who actually defaulted ($y_i = 1$).
- ▶ Second Product ($\prod_{i:y_i=0} (1 - p(x_i))$): Multiplies the predicted probabilities of not defaulting ($1 - p(x_i)$) for all individuals who did not default ($y_i = 0$).
- ▶ **Meaning**: The likelihood function represents the joint probability of observing the specific set of zeros and ones present in the dataset.

Maximum Likelihood –continued

We use **maximum likelihood** to estimate the parameters:

$$\ell(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i:y_i=0} (1 - p(x_i)).$$

- ▶ This likelihood gives the probability of the observed zeros and ones in the data. We pick β_0 and β_1 to maximize the likelihood of the observed data.
- ▶ In R we use the `glm` function with `family = binomial`.

	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-10.6513	0.3612	-29.5	< 0.0001
balance	0.0055	0.0002	24.9	< 0.0001

Hypothesis Testing: z -statistic vs. t -statistic

Linear Regression (Chapter 3)

- ▶ **Fitting Method:** Ordinary Least Squares (OLS).
- ▶ **Error Variance (σ^2):** Unknown and must be estimated from data via Residual Standard Error (RSE).
- ▶ **Exact Distribution:** Under normality of errors, dividing by the estimated SE yields an exact **Student's t -distribution**:

$$t = \frac{\hat{\beta}_j - 0}{\text{SE}(\hat{\beta}_j)} \sim t_{n-p-1}$$

Logistic Regression (Chapter 4)

- ▶ **Fitting Method:** Maximum Likelihood Estimation (MLE).
- ▶ **Variance Determined:** For binary outcomes, $\text{Var}(Y|X) = p(X)(1 - p(X))$; there is no separate σ^2 parameter.
- ▶ **Asymptotic Normality:** By MLE theory (large samples), the Wald statistic follows a **standard normal distribution**:

$$z = \frac{\hat{\beta}_j - 0}{\text{SE}(\hat{\beta}_j)} \sim \mathcal{N}(0, 1)$$

Making Predictions

- What is our estimated probability of default for someone with a balance of \$1000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}} = \frac{e^{-10.6513 + 0.0055 \times 1000}}{1 + e^{-10.6513 + 0.0055 \times 1000}} = 0.006$$

Making Predictions

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- With a balance of \$2000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}} = \frac{e^{-10.6513 + 0.0055 \times 2000}}{1 + e^{-10.6513 + 0.0055 \times 2000}} = 0.586$$

Using `student` as the Predictor

Let's fit a logistic regression model with `student` as the single predictor:

	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-3.5041	0.0707	-49.55	< 0.0001
student[Yes]	0.4049	0.1150	3.52	0.0004

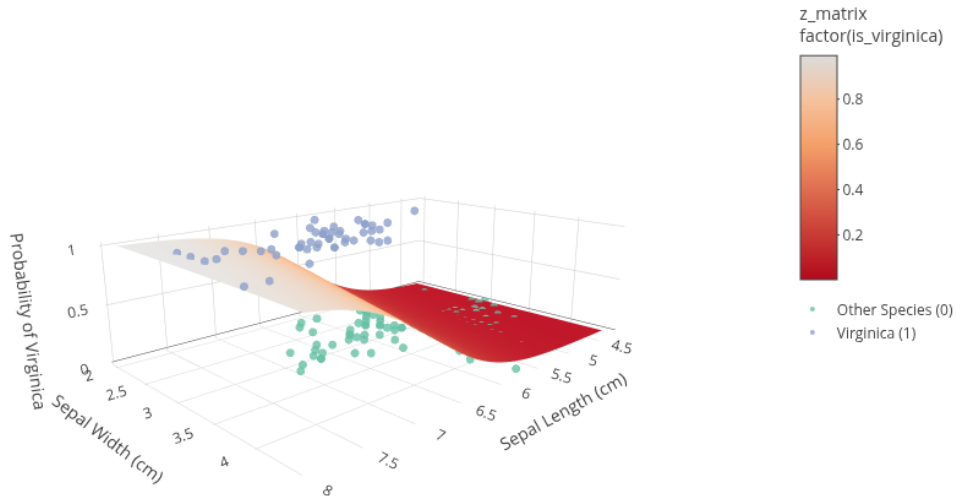
Predicted probabilities:

$$\widehat{\Pr}(\text{default} = \text{Yes} \mid \text{student} = \text{Yes}) = \frac{e^{-3.5041+0.4049 \times 1}}{1 + e^{-3.5041+0.4049 \times 1}} = 0.0431,$$

$$\widehat{\Pr}(\text{default} = \text{Yes} \mid \text{student} = \text{No}) = \frac{e^{-3.5041+0.4049 \times 0}}{1 + e^{-3.5041+0.4049 \times 0}} = 0.0292.$$

Logistic Regression with Several Variables

3D Logistic Regression Surface (Iris Dataset)



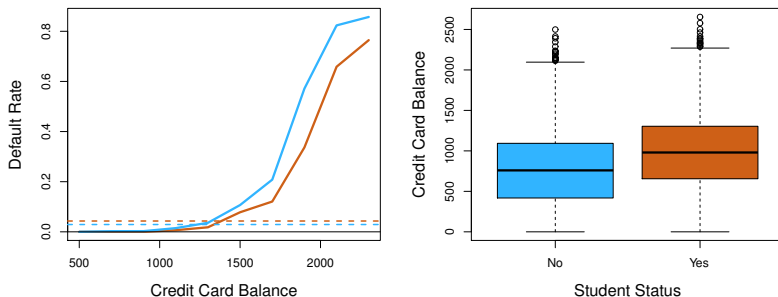
Logistic Regression with Several Variables –continued

$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p \implies p(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}$$

	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-10.8690	0.4923	-22.08	< 0.0001
balance	0.0057	0.0002	24.74	< 0.0001
income	0.0030	0.0082	0.37	0.7115
student[Yes]	-0.6468	0.2362	-2.74	0.0062

*Why is the coefficient for **student** negative now, while it was positive in the simple model?*

Confounding in the Credit Card Data



Confounding Phenomenon (Simpson's Paradox): Left (**Default Rate vs. Balance**): For any given credit card balance level, students (orange line) have a lower default rate than non-students (blue line). Right (**Student vs. Balance Boxplot**): However, students tend to hold substantially higher credit card balances on average compared to non-students.

- ▶ Students have higher overall (marginal) default rate because they carry more debt.
- ▶ But controlling for balance, being a student is associated with lower default risk.
- ▶ Multiple logistic regression teases this out!

The Apparent Contradiction: Marginal vs. Conditional

- **Scenario:** We want to predict whether an individual will **Default** on credit card debt using **Student** status (1 = Yes, 0 = No) and **Balance**.

Simple Logistic Regression (Marginal)

$$\Pr(\text{Def} = \text{Yes}) = \frac{e^{\beta_0 + \beta_1 \cdot \text{Stu}}}{1 + e^{\beta_0 + \beta_1 \cdot \text{Stu}}}$$

- $\hat{\beta}_1 = +0.405$ ($p < 0.001$, positive)
- **Interpretation:** Students are **more likely** to default overall than non-students (4.3% vs 2.9%).

Multiple Logistic Regression (Conditional)

$$\Pr(\text{Def} = \text{Yes}) = \frac{e^{\beta_0 + \beta_1 \text{Stu} + \beta_2 \text{Bal} + \beta_3 \text{Inc}}}{1 + e^{\dots}}$$

- $\hat{\beta}_1 = -0.647$ ($p = 0.006$, negative!)
- **Interpretation:** For a **fixed balance**, students are **less likely** to default.

The Core Question

Why does the sign of $\hat{\beta}_{\text{Student}}$ flip from **positive** to **negative**?

The Mechanism: Confounding via Debt Balance

The reversal of sign is driven by the relationship between the predictor variables:

1. Balance is the Dominant Risk Factor:

- ▶ Higher credit card balance dramatically increases the probability of default ($\hat{\beta}_{\text{Balance}} > 0$).

2. Correlation Between Student Status and Balance:

- ▶ Students systematically carry **higher average balances** than non-students.

3. Holding Balance Fixed (Conditional Comparison):

- ▶ When comparing a student and a non-student with the *same* balance, the student's probability of default is actually lower.

Confounding / Simpson's Paradox Summary

In simple regression, **Student** absorbs the omitted effect of **Balance**. Once **Balance** is controlled for in multiple regression, the true conditional effect of **Student** is revealed.

Practical Takeaways for Credit Scoring & Inference

Decision-Making Scenarios

- ▶ **Balance is Unknown:** Students are riskier applicants on average $\Rightarrow$ issue lower credit or higher interest.
- ▶ **Balance is Known:** A student with balance $\$b$ is **less risky** than a non-student with the same balance $\$b$.

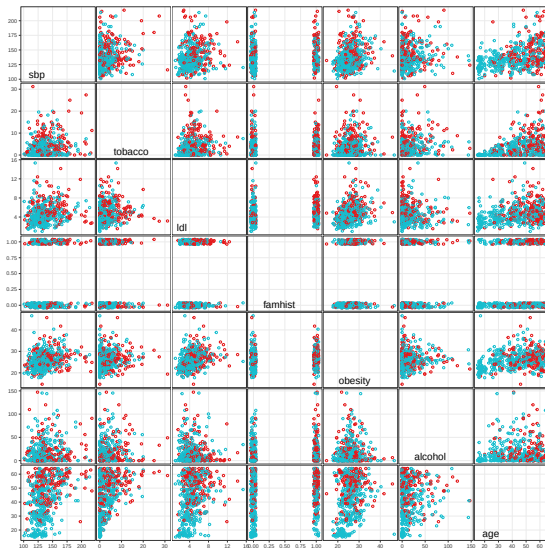
Key Statistical Lessons

- ▶ **Omitted Variable Bias:** Marginal associations can be severely misleading when key confounders are omitted.
- ▶ **Multivariate Modeling:** Coefficients represent effects *holding other variables constant*.
- ▶ **Correlation $\neq$ Causation:** Always consider the joint distribution of predictors.

Example: South African Heart Disease

- ▶ 160 cases of MI (myocardial infarction) and 302 controls (all male in age range 15–64), from Western Cape, South Africa in the early 1980s.
- ▶ Overall prevalence is very high in this region: 5.1%.
- ▶ Measurements on seven predictors (risk factors): **sbp** (systolic blood pressure), **tobacco**, **ldl** (low density lipoprotein cholesterol), **famhist** (family history of heart disease), **obesity**, **alcohol**, and **age**.
- ▶ Goal is to identify relative strengths and directions of risk factors.
- ▶ Part of an intervention study aimed at educating the public on healthier diets.

South African Heart Disease: Scatterplot Matrix



Scatterplot matrix of the South African Heart Disease data. The response is color coded —the cases (MI) are red, and the controls are turquoise. `famhist` is a binary variable, with 1 indicating family history of MI.

MI: myocardial infarction;
sbp: systolic blood pressure;
ldl: low-density lipoprotein;
chd: coronary heart disease.

SA Heart Disease: Logistic Regression Model Output

```
> heartfit<-glm(chd~.,data=heart,family=binomial)
> summary(heartfit)
```

Call:

```
glm(formula = chd ~ ., family = binomial, data = heart)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-4.1295997	0.9641558	-4.283	1.84e-05	***
sbp	0.0057607	0.0056326	1.023	0.30643	
tobacco	0.0795256	0.0262150	3.034	0.00242	**
ldl	0.1847793	0.0574115	3.219	0.00129	**
famhistPresent	0.9391855	0.2248691	4.177	2.96e-05	***
obesity	-0.0345434	0.0291053	-1.187	0.23529	
alcohol	0.0006065	0.0044550	0.136	0.89171	
age	0.0425412	0.0101749	4.181	2.90e-05	***

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 596.11 on 461 degrees of freedom
Residual deviance: 483.17 on 454 degrees of freedom
AIC: 499.17

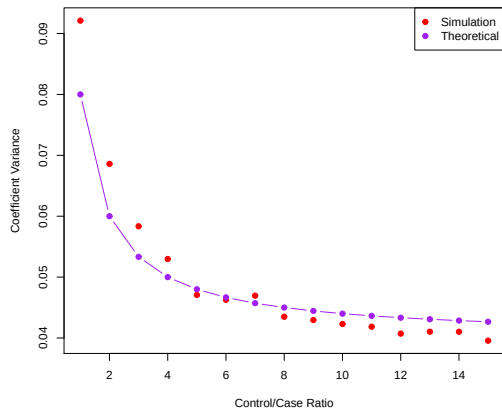
Case-Control Sampling and Logistic Regression

- ▶ In the SA data, there are 160 cases and 302 controls ($\tilde{\pi} = 0.35$ cases). Yet the true population prevalence is $\pi = 0.05$.
- ▶ With case-control samples, we can estimate slope parameters β_j accurately (if our model is correct); only the constant term β_0 is biased.
- ▶ We can correct the estimated intercept by:

$$\hat{\beta}_0^* = \hat{\beta}_0 + \log\left(\frac{\pi}{1-\pi}\right) - \log\left(\frac{\tilde{\pi}}{1-\tilde{\pi}}\right)$$

- ▶ Often cases are rare and we take all of them; up to 5 times that number of controls is sufficient.

Diminishing Returns in Unbalanced Binary Data



Control/Case Ratio vs. Coefficient Variance: As the control-to-case sampling ratio increases from 1 to 14, the variance of the parameter estimates decreases rapidly up to a ratio of about 5 : 1, after which the variance reduction flattens out.

Diminishing Returns in Unbalanced Binary Data –continued

Motivation & The Case-Control Design

► The Rare Outcome Challenge:

- In medical studies (e.g., myocardial infarction), the outcome is rare in the general population ($\pi = \Pr(Y = 1) \approx 5\%$).
- A simple random sample requires massive sample sizes to observe enough positive cases.

► Case-Control Strategy:

- Collect **all** available cases ($Y = 1$) and sample a manageable group of controls ($Y = 0$).

► Proportion Discrepancy (e.g., SA Heart Disease Study):

- Population Prevalence (π): ≈ 0.05 (5.1% true risk)
- Sample Proportion ($\tilde{\pi}$): $\frac{160}{160+302} \approx 0.35$ (35% in sample)

► Key Question: Does artificially inflating cases distort parameter estimation?

Diminishing Returns in Unbalanced Binary Data –continued

Invariance of Slope Estimates

► Log-Odds Representation:

$$\log \left(\frac{\Pr(Y = 1 \mid X)}{1 - \Pr(Y = 1 \mid X)} \right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

► Mathematical Invariance Property:

- Under case-control sampling, the relationship between features X and the outcome Y is preserved.
- The estimated slope coefficients $\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_p$ are **unbiased** and match population log-odds ratios.

► The Catch:

- Only the intercept term β_0 is biased upward because it absorbs the inflated baseline rate of cases.

Diminishing Returns in Unbalanced Binary Data –continued

Intercept Correction for True Probabilities

► Why Correct the Intercept?

- Unadjusted predictions overestimate the true risk of disease in the population.

► Correction Formula (when true prevalence π is known):

$$\hat{\beta}_0^* = \hat{\beta}_0 + \log\left(\frac{\pi}{1-\pi}\right) - \log\left(\frac{\tilde{\pi}}{1-\tilde{\pi}}\right)$$

- $\log\left(\frac{\pi}{1-\pi}\right)$: Log-odds of the disease in the **population**.
- $\log\left(\frac{\tilde{\pi}}{1-\tilde{\pi}}\right)$: Log-odds of the disease in the **case-control sample**.

► Calibrated Population Posterior Probabilities:

$$\widehat{\Pr}(Y = 1 \mid X) = \frac{e^{\hat{\beta}_0^* + \sum_{j=1}^p \hat{\beta}_j X_j}}{1 + e^{\hat{\beta}_0^* + \sum_{j=1}^p \hat{\beta}_j X_j}}$$

Diminishing Returns in Unbalanced Binary Data –continued

Practical Takeaways & The 5:1 Sampling Rule

Key Takeaways:

- ▶ **Sample Efficiency:** For rare outcomes, case-control sampling dramatically cuts data collection costs without sacrificing slope accuracy.
- ▶ **Diminishing Returns:** Increasing the control-to-case ratio beyond $\approx 5 : 1$ yields negligible variance reduction.
- ▶ **Classification vs. Odds:** Odds ratios are valid immediately; correct β_0 only when absolute probability is needed.

Control-to-Case Ratio

- ▶ **1:1 to 4:1:** Steep decrease in parameter variance $\text{Var}(\hat{\beta}_j)$.
- ▶ $\geq$ **5:1:** Variance curve flattens out.
- ▶ **Rule of Thumb:** Up to 5 controls per case is sufficient in practice.

Logistic Regression with More than Two Classes

- ▶ Generalizes naturally to $K > 2$ classes (Multinomial Logistic Regression).
- ▶ Symmetric (softmax) formulation:

$$\Pr(Y = k \mid X) = \frac{e^{\beta_{0k} + \beta_{1k}X_1 + \dots + \beta_{pk}X_p}}{\sum_{\ell=1}^K e^{\beta_{0\ell} + \beta_{1\ell}X_1 + \dots + \beta_{p\ell}X_p}}$$

- ▶ Here there is a linear function for **each** class.
- ▶ Redundancy: only $K - 1$ linear functions are required (one class serves as baseline).
- ▶ Multiclass logistic regression is also referred to as **multinomial regression**.

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Discriminant Analysis

- ▶ **Generative approach:** Model the distribution of X in each of the classes separately ($f_k(x)$), and then use **Bayes' theorem** to flip things around to obtain $\Pr(Y | X)$.
- ▶ When we use normal (Gaussian) distributions for each class, this leads to linear or quadratic discriminant analysis according to the covariance matrix:
 - ▶ Common $\Sigma \implies$ **Linear Discriminant Analysis (LDA)**.
 - ▶ Class-specific $\Sigma_k \implies$ **Quadratic Discriminant Analysis (QDA)**.
- ▶ However, this approach is quite general, and other distributions can be used as well. We will focus on normal distributions.

Bayes' Theorem for Classification

- ▶ Thomas Bayes was a famous mathematician whose name represents a big subfield of statistical and probabilistic modeling. Here we focus on a simple result, known as Bayes theorem:

$$\Pr(Y = k \mid X = x) = \frac{\Pr(X = x \mid Y = k) \cdot \Pr(Y = k)}{\Pr(X = x)}$$

- ▶ For discriminant analysis, this is written as:

$$p_k(x) = \Pr(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{\ell=1}^K \pi_\ell f_\ell(x)}$$

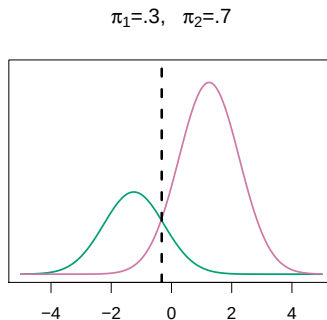
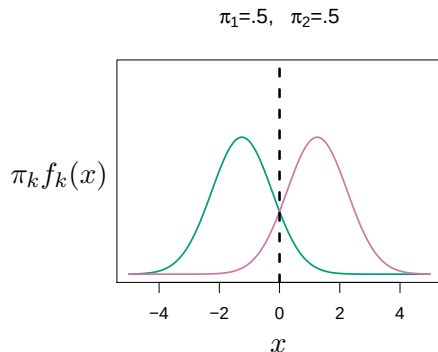
where:

- ▶ $f_k(x) = \Pr(X = x \mid Y = k)$ is the **class-conditional density** for X in class k .
- ▶ $\pi_k = \Pr(Y = k)$ is the marginal or **prior probability** for class k .

Classify to the Highest Density

1D Densities and Bayes Decision Boundary:

- ▶ **Equal Priors** ($\pi_1 = 0.5, \pi_2 = 0.5$): Decision boundary is at the point where the two density curves intersect ($x = \frac{\mu_1 + \mu_2}{2}$).
- ▶ **Unequal Priors** ($\pi_1 = 0.3, \pi_2 = 0.7$): The decision boundary shifts toward class 1 (left) to favor the higher prior probability of class 2.



Why Discriminant Analysis?

- ▶ **Separation problem:** When classes are well-separated, logistic regression parameter estimates are surprisingly unstable (coefficients diverge to $\pm\infty$). LDA does not suffer from this.
- ▶ **Small sample size:** If n is small and the distribution of predictors is approximately Gaussian in each class, LDA is more stable and efficient than logistic regression.
- ▶ **Multiclass ($K > 2$):** LDA provides convenient low-dimensional visualizations and projections of the data.

Linear Discriminant Analysis when $p = 1$

The Gaussian (normal) density has the form:

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} e^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma_k}\right)^2}$$

Here μ_k is the mean, and σ_k^2 the variance (in class k). We will assume that all the $\sigma_k = \sigma$ are the same ($\sigma_1^2 = \dots = \sigma_k^2 = \sigma^2$).

¹Shared variance across all classes is the foundational assumption that defines LDA in order to cancel quadratic terms.

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So, Bayes' formula gives us a rather complex expression for $p_k(x) = Pr(Y = k|X = x)$:

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma_k}\right)^2}}{\sum_{\ell=1}^K \pi_\ell \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu_\ell}{\sigma_\ell}\right)^2}}$$

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Linear Discriminant Analysis when $p = 1$

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Happily, there are simplifications and cancellations¹.

¹Shared variance across all classes is the foundational assumption that defines LDA in order to cancel quadratic terms.

Discriminant Functions ($p = 1$)

Taking logs and removing common constants, maximizing $p_k(x)$ is equivalent to maximizing the **linear discriminant score**:

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

- Note that $\delta_k(x)$ is a **linear** function of x .

Discriminant Functions ($p = 1$)

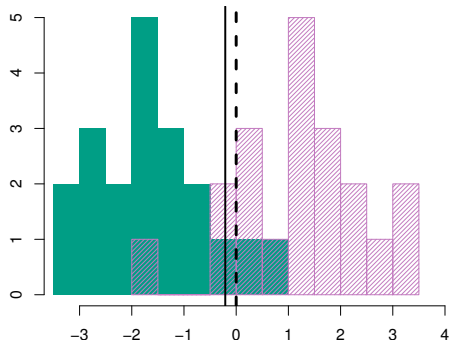
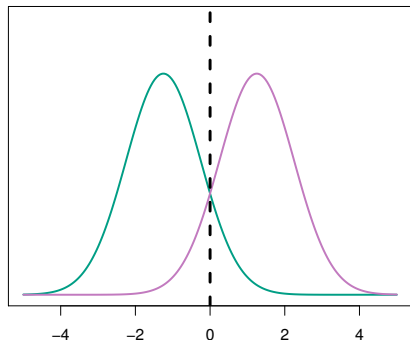
Taking logs and removing common constants, maximizing $p_k(x)$ is equivalent to maximizing the **linear discriminant score**:

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

- ▶ Note that $\delta_k(x)$ is a **linear** function of x .
- ▶ For $K = 2$ classes and equal priors ($\pi_1 = \pi_2 = 0.5$), the **decision boundary** ($\delta_1(x) = \delta_2(x)$) is:

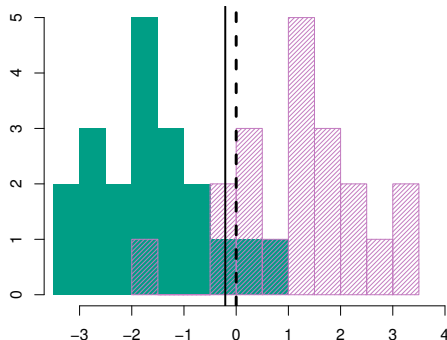
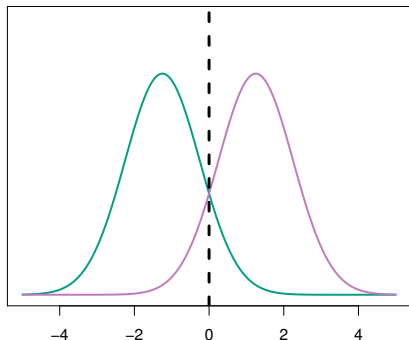
$$x = \frac{\mu_1 + \mu_2}{2}$$

Discriminant Functions ($p = 1$) –continued



- Example with $\mu_1 = -1.5$, $\mu_2 = 1.5$, $\pi_1 = \pi_2 = 0.5$, and $\sigma^2 = 1$.

Discriminant Functions ($p = 1$) –continued



- ▶ Example with $\mu_1 = -1.5$, $\mu_2 = 1.5$, $\pi_1 = \pi_2 = 0.5$, and $\sigma^2 = 1$.
- ▶ Typically we don't know these parameters; we just have the training data. In that case we simply estimate the parameters and plug them into the rule.

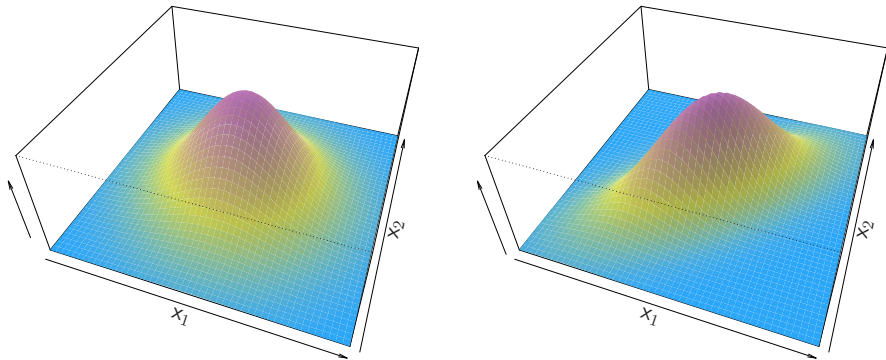
Estimating the Parameters for LDA

In practice, true parameters are unknown and estimated from training data:

$$\begin{aligned}\hat{\pi}_k &= \frac{n_k}{n} \\ \hat{\mu}_k &= \frac{1}{n_k} \sum_{i:y_i=k} x_i \\ \hat{\sigma}^2 &= \frac{1}{n-K} \sum_{k=1}^K \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2 = \sum_{k=1}^K \frac{n_k - 1}{n - K} \hat{\sigma}_k^2\end{aligned}$$

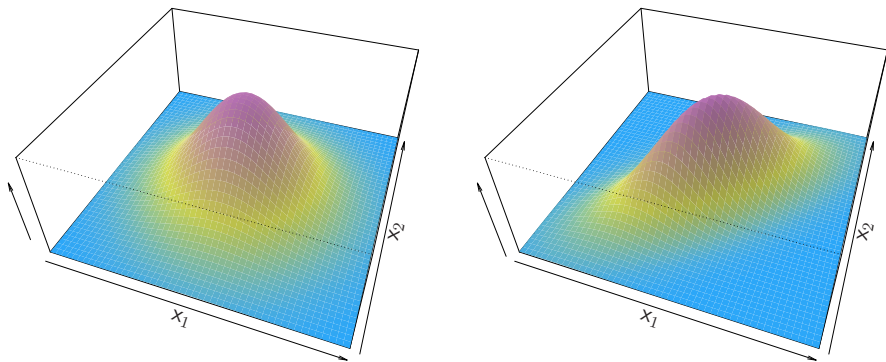
where $\hat{\sigma}_k^2 = \frac{1}{n_k-1} \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2$ is the usual formula for the estimated variance in the class k .

Linear Discriminant Analysis when $p > 1$



$$\text{Density: } f(x) = \frac{1}{(2\pi)^{\frac{p}{2}} |\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

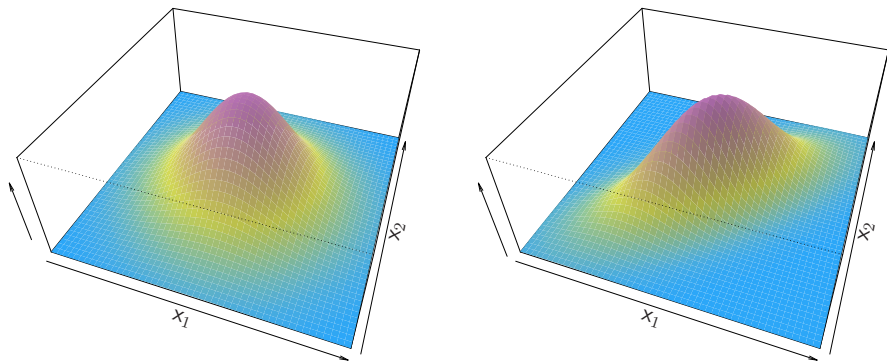
Linear Discriminant Analysis when $p > 1$



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► Discriminant function: $\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k$

Linear Discriminant Analysis when $p > 1$



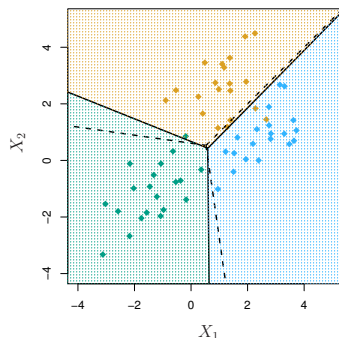
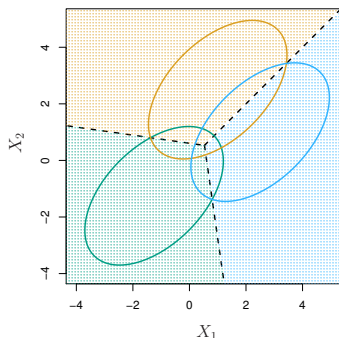
$$\text{Density: } f(x) = \frac{1}{(2\pi)^{\frac{p}{2}} |\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

- ▶ Discriminant function: $\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k$
- ▶ Despite its complex form, $\delta_k(x) = c_{k0} + c_{k1}x_1 + c_{k2}x_2 + \cdots + c_{kp}x_p$ is a **linear function** of x .

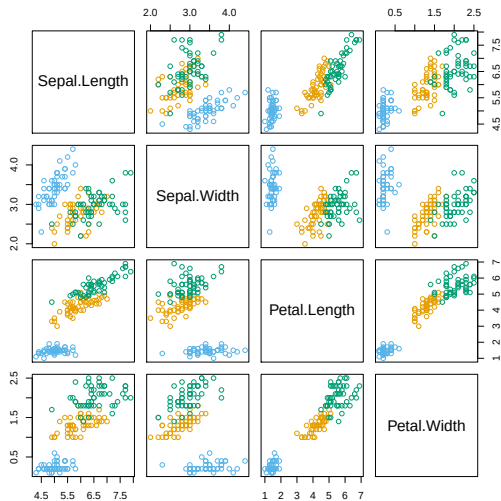
Illustration: $p = 2$ and $K = 3$ Classes

2D Gaussian LDA Contours and Decision Boundaries:

- ▶ Here we have 3 classes with equal priors ($\pi_1 = \pi_2 = \pi_3 = \frac{1}{3}$).
- ▶ Elliptical contour lines represent bivariate Gaussian densities with shared covariance Σ .
- ▶ Dashed lines represent optimal **Bayes decision boundaries** dividing the plane into three polygonal regions.
- ▶ Were they known, they would yield the fewest misclassification errors, among all possible classifiers.

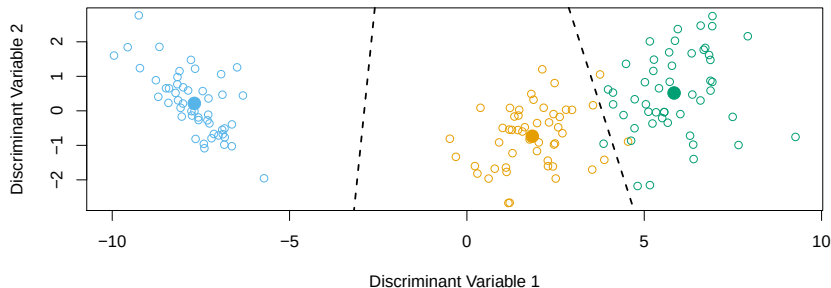


Fisher's Iris Data



- ▶ 4 continuous measurements: Sepal.Length, Sepal.Width, Petal.Length, Petal.Width.
- ▶ 3 species ($K = 3$): *Setosa* (blue), *Versicolor* (orange), *Virginica* (green) (50 samples per class, $n = 150$).
- ▶ LDA correctly classifies all but 3 of the 150 training samples (2% training error).

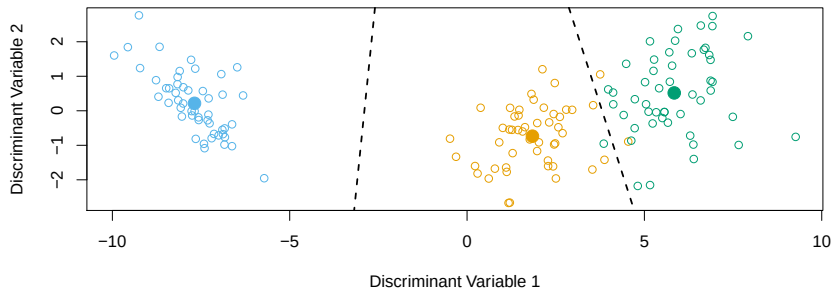
Fisher's Discriminant Plot



Fisher's Linear Discriminants for Iris Data:

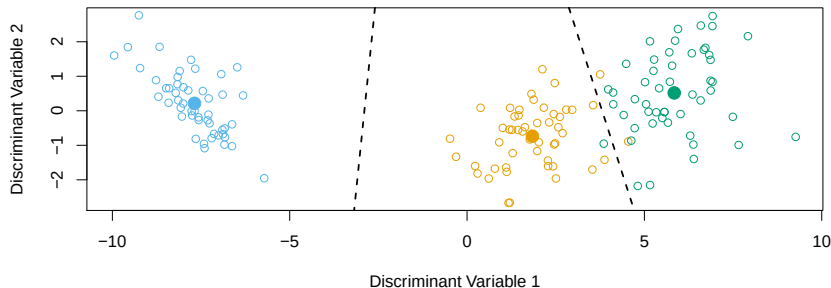
- ▶ When there are K classes, linear discriminant analysis can be viewed exactly in a $K - 1$ dimensional plot.
- ▶ Why?

Fisher's Discriminant Plot



- ▶ When there are K classes, linear discriminant analysis can be viewed exactly in a $K - 1$ dimensional plot.
- ▶ **Why?** Because LDA essentially classifies to the closest centroid in Mahalanobis distance, and K centroids span at most a $(K - 1)$ -dimensional hyperplane.

Fisher's Discriminant Plot



- ▶ When there are K classes, linear discriminant analysis can be viewed exactly in a $K - 1$ dimensional plot.
- ▶ **Why?** Because LDA essentially classifies to the closest centroid in Mahalanobis distance, and K centroids span at most a $(K - 1)$ -dimensional hyperplane.
- ▶ Even when $K > 3$, we can find the “best” 2-dimensional plane for visualizing the discriminant rule.

From $\delta_k(x)$ to Class Probabilities

Once we have estimates $\hat{\delta}_k(x)$, we convert them into posterior class probabilities:

$$\widehat{\Pr}(Y = k \mid X = x) = \frac{e^{\hat{\delta}_k(x)}}{\sum_{\ell=1}^K e^{\hat{\delta}_\ell(x)}}$$

- ▶ Classifying to the largest $\hat{\delta}_k(x)$ is identical to classifying to the class with highest posterior probability.
- ▶ For $K = 2$, we predict class 2 if $\widehat{\Pr}(Y = 2 \mid X = x) \geq 0.5$, else class 1.

Plan

An Overview of Classification

Logistic Regression

Linear Discriminant Analysis

Classification Evaluation & Other Classifiers

LDA on Credit Data: Confusion Matrix

		True Default Status		Total
		No	Yes	
Predicted Default Status	No	9644	252	9896
	Yes	23	81	104
Total		9667	333	10000

$\frac{23+252}{10000} = \frac{275}{10000}$ errors –an apparent *misclassification rate* of 2.75%.

LDA on Credit Data: Confusion Matrix

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- ▶ A trivial **null classifier** (always predicting No) has an error rate of $333/10000 = 3.33\%$.
- ▶ **False Positive Rate (True No's errors):** $\frac{23}{9667} = 0.2\%$.
- ▶ **False Negative Rate (True Yes's errors):** $\frac{252}{333} = 75.7\%$ (missed 75.7% of defaults!).

Types of Errors & Varying the Threshold

False positive rate:

The fraction of negative examples that are classified as positive $\rightarrow$ 0.2% in example.

False negative rate:

The fraction of positive examples that are classified as negative $\rightarrow$ 75.7% in example.

Types of Errors & Varying the Threshold

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The fraction of negative examples that are classified as positive $\rightarrow$ 0.2% in example.

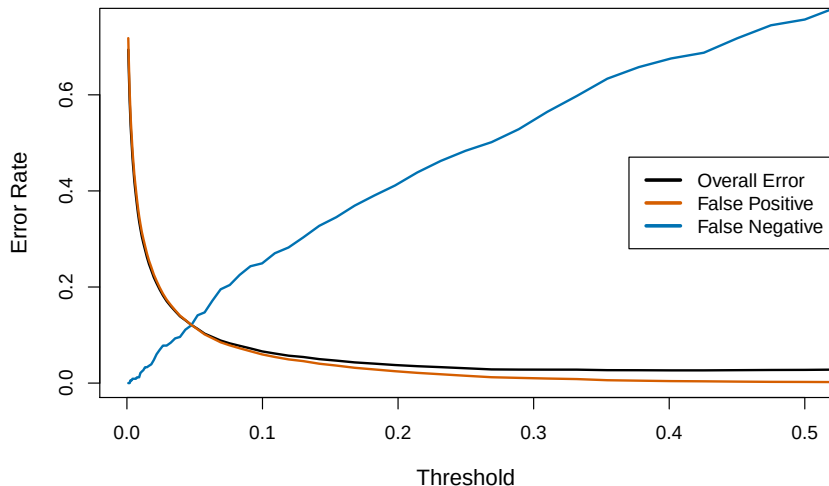
False negative rate:

The fraction of positive examples that are classified as negative $\rightarrow$ 75.7% in example.

- ▶ Defaulting threshold was set at 0.5: predict default if $\widehat{\Pr}(\text{Default} = \text{Yes} \mid X) \geq 0.5$.
- ▶ In credit card default, a false negative is much more costly than a false positive.
- ▶ We can vary the threshold to any value in $[0, 1]$:

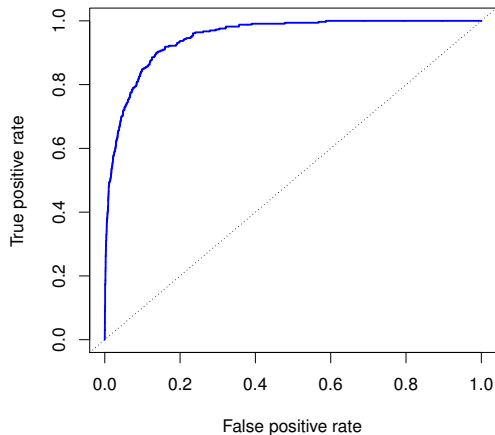
$$\widehat{\Pr}(\text{Default} = \text{Yes} \mid X) \geq \textit{threshold}.$$

Types of Errors & Varying the Threshold –continued



Threshold vs. Error Rates: Lowering threshold from 0.5 down to 0.1 dramatically drops the false negative rate with only a modest increase in false positives and overall error.

ROC Curve



Receiver Operating Characteristic (ROC) Curve:

- ▶ Displays **True Positive Rate** (Sensitivity) vs. **False Positive Rate** ($1 - \text{Specificity}$) across all possible classification thresholds.
- ▶ An ideal ROC curve hugs the top-left corner.
- ▶ The performance is summarized by the **AUC** (Area Under the ROC Curve).
- ▶ AUC ranges from 0.5 (random guessing) to 1.0 (perfect classification). Higher AUC is better.

Other Forms of Discriminant Analysis

Recall Bayes' formula:

$$\Pr(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{\ell=1}^K \pi_\ell f_\ell(x)}$$

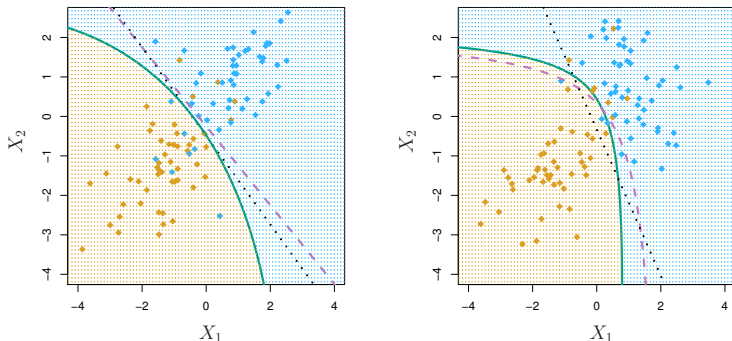
When $f_k(x)$ are Gaussian densities, with the same covariance matrix Σ in each class, this leads to linear discriminant analysis. By altering the forms for $f_k(x)$, we get different classifiers.

- ▶ **LDA:** $f_k(x)$ is Gaussian with shared covariance matrix Σ .
- ▶ **QDA (Quadratic DA):** $f_k(x)$ is Gaussian with **class-specific** covariance matrices Σ_k .
- ▶ **Naive Bayes:** $f_k(x) = \prod_{j=1}^p f_{kj}(x_j)$ assumes conditional independence across features within each class (for Gaussian this means the Σ_k are diagonal).
- ▶ **Non-parametric DA:** Uses kernel density estimation or histograms for $f_k(x)$.

Quadratic Discriminant Analysis (QDA)

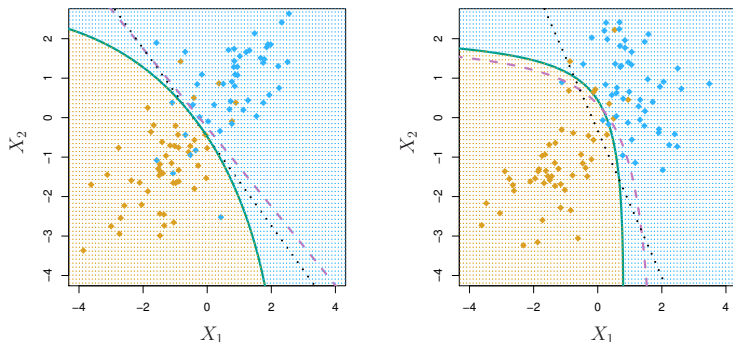
When Σ_k are distinct for each class k , the quadratic terms no longer cancel:

$$\delta_k(x) = -\frac{1}{2}(x - \mu_k)^T \Sigma_k^{-1}(x - \mu_k) - \frac{1}{2} \log |\Sigma_k| + \log \pi_k$$



QDA vs. LDA Decision Boundaries:

Quadratic Discriminant Analysis (QDA) –continued



QDA vs. LDA Decision Boundaries:

- ▶ When classes have different correlation structures/variances, the Bayes boundary is curved (quadratic).
- ▶ QDA fits curved quadratic boundaries, capturing complex interactions at the cost of estimating $K \times p(p+1)/2$ covariance parameters.

Naive Bayes

- ▶ Assumes features are conditionally independent in each class:
 $f_k(x) = \prod_{j=1}^p f_{kj}(x_j).$
- ▶ Useful when p is large (where covariance estimation in LDA/QDA breaks down).
- ▶ **Gaussian Naive Bayes:** Σ_k is diagonal (σ_{kj}^2):

$$\delta_k(x) \propto \log \left[\pi_k \prod_{j=1}^p f_{kj}(x_j) \right] = -\frac{1}{2} \sum_{j=1}^p \left[\frac{(x_j - \mu_{kj})^2}{\sigma_{kj}^2} + \log \sigma_{kj}^2 \right] + \log \pi_k$$

- ▶ **Mixed features:** Easily handles qualitative features using discrete probability tables/histograms.
- ▶ Despite strong independence assumptions, Naive Bayes often performs surprisingly well.

Logistic Regression versus LDA

For a two-class problem, the log-odds for LDA is:

$$\log \left(\frac{p_1(x)}{1 - p_1(x)} \right) = \log \left(\frac{p_1(x)}{p_2(x)} \right) = c_0 + c_1 x_1 + \cdots + c_p x_p$$

Both models produce linear decision boundaries! The key difference lies in how the parameters are estimated:

²Logistic regression can fit non-linear boundaries by explicitly adding polynomial or interaction terms.

Logistic Regression versus LDA

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Both models produce linear decision boundaries! The key difference lies in how the parameters are estimated:

- ▶ **Logistic Regression² (Discriminative learning):** Maximizes conditional likelihood $\Pr(Y | X)$. Makes fewer distributional assumptions.
- ▶ **LDA (Generative learning):** Maximizes full joint likelihood $\Pr(X, Y)$ assuming Gaussianity. More efficient when normality holds.
- ▶ Despite these differences, in practice the results are often very similar.

²Logistic regression can fit non-linear boundaries by explicitly adding polynomial or interaction terms.

Summary of Classification Methods

- ▶ **Logistic Regression:** Very popular and robust for binary ($K = 2$) and multiclass problems.
- ▶ **LDA:** Well-suited when n is small, classes are well-separated, or normality holds; provides dimension reduction.
- ▶ **QDA:** Useful when sample size is large and covariance matrices differ noticeably between classes.
- ▶ **Naive Bayes:** Ideal when p is very large or features are mixed categorical/continuous.
- ▶ **KNN:** Non-parametric, flexible when decision boundary is highly irregular, but suffers in high dimensions (Section 4.5).